Data Engineering [D.E]
Assignment - 04

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7/ Describe the raw ingredients of data storage in modern data systems.

Raw ingredients of data storage: These are the basic components that make modern ~~sy~~ data systems work efficiently. these ingredients supports how data is stored processed and retreived in data architectures like data warehouse and data lakes.

Let us look at some of the raw ingredients of data storage:

Magnetic disk drive: Magnetic disks utilize spinning platters coated with a ferromagnetic film. This flim is magnetized by a read/write head during write operations to physically encode binary data. Magnetic disks drives have been around for ages. They act as backbone of bulk data storage systems because they are cheaper than the SSDs per gigabyte of stored data.

Solid-state Drive: Solid-sdstate drives (SSDs) store data as charges in flash memory cells. SSDs eliminate the mechanical components of magnetic drives, the data is stead by purely electronic means. SSDs can look up random data in less than 0.1 ms (100 microseconds).

Random Access Memory: RAM has several specific characteristics:

- It is attached to a CPU & mapped into CPU address space
- It stores the code that CPUs execute
- It is volatile, while magnetic drives & SSDs are nonvolatile
- It offers significantly higher transfer speeds and faster retrieval times than SSD storage.

It is significantly more expensive than SSD storage.
Networking and CPU: CPU handle the details of servicing, requests, aggregating reads & distributing writes.
Storage becomes a web application with an API network device performance and network topology are key factors in realizing high performance.
Serialization: Serialization is another raw storage ingredient & a critical element of database design. The decisions around serialization will inform how well queries perform across a network, CPU overhead, query latency and more.
Compression: It is the process of reducing the size of data so that it takes less storage space. In modern data systems, Compression is a very important component because large amounts of data are stored, processed and transferred between systems.
Caching: It is a technique used in storage systems where frequently or recently used data is stored in a fast storage layer so it can be accessed quickly.

8) Discuss different storage models and their impact on data retrieval, transformation & analysis.

Ans: The different storage models determine how data is organized and stored which directly affects data retrieval, transformation & analysis in modern data systems. The main storage models discussed in data engineering are:

1) Row-oriented storage: Data is stored row by row.
- Fast for transaction processing
- Good for retrieving complete records
- Less efficient for large analytical queries.

2) Column-oriented storage: Data is stored column by column.
- Fast for analytical queries & aggregation
- Reads only required columns
- Improves query performance & compression

3) Key-value storage: Stores data as key-value pairs
- Very fast retrieval using keys
- Limited query & transformation support
- used in caching and real-time systems.

4) Document storage: stores data in document format like JSON
- Flexible schema
- Easy data transformation
- Suitable for semi-structured data.

5) Graph storage: stores data as nodes & relationships
- Efficient for relationship analysis
- Useful for social networks and recommendation systems.
- Supports complex queries.

9) Explain in detail the following while ingesting events: a) Schema evolution b) late arriving data c) ordering and multiple delivery d) Replay e) Time to live.

During event ingestion, systems must handle schema changes, delayed data, event ordering issues, duplicate deliveries, the ability to replay data, and data retention limits (TTL)

1. Schema Evolution: changes in the data structure over time while the ingestion system continues processing old and new data formats.

2. Late Arriving data: Data that reaches the system after the expected processing time due to delays in networks or systems.

3. Ordering and Multiple delivery:- Events may arrive out of order or may be delivered more than once in distributed systems.

4. Replay: The ability to reprocess previously stored events to fix errors or rebuild the data pipelines

5. Time to Live (TTL): The time period for which data is stored in the system before it is automatically deleted.

proper management of these factors ensures reliable, scalable and accurate data pipelines in modern data engineering systems.